Linear Algebra and Its Applications

Section 1.9: The Matrix of a Linear Transformation

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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1.9 The matrix of a linear transformation

For a linear map $T : \mathbb{R}^n \to \mathbb{R}^m$, its standard matrix is

$$A = [T(e_1) \ \cdots \ T(e_n)].$$

The image of e_j is column j, not row j. This works because $x = \sum_j x_j e_j$ and linearity gives $T(x) = \sum_j x_j T(e_j) = Ax$. The map is one-to-one exactly when every column has a pivot, so $Ax = 0$ has only the trivial solution. It is onto $\mathbb{R}^m$ exactly when every row has a pivot. One-to-one means at most one input for any output; onto means at least one input for every output in the codomain. For a composition, the transformation applied first corresponds to the matrix farther to the right in the product.

Practice problem 1.9.P1: A shear followed by reflection

The shear fixes e_1 and takes e_2 to $e_2 - \frac{1}{2}e_1$. Reflection across the x_2-axis negates the first coordinate and leaves the second unchanged. Thus the final images are $T(e_1) = (-1, 0)^T$ and $T(e_2) = (1/2, 1)^T$. Therefore

$$A = \begin{bmatrix} -1 & 1/2 \\ 0 & 1 \end{bmatrix}.$$

We follow the image through both operations before making it a column.

Exercise 1.9.1: Put two four-coordinate images into columns

The domain has dimension two and the output has four coordinates. Use the two specified images in their stated order:

$$A = [T(e_1) \ T(e_2)] = \begin{bmatrix} 3 & -5 \\ 1 & 2 \\ 3 & 0 \\ 1 & 0 \end{bmatrix}.$$

Indeed its first column is $T(e_1)$ and its second is $T(e_2)$, which is enough to determine the linear map on every input.

Exercise 1.9.2: Three input coordinates and two output coordinates

There must be three columns and two rows. The standard matrix is

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 9 & -8 \end{bmatrix}.$$

For example, the second column is the given image $(-2, 9)^T$. Multiplying A by any $(x_1, x_2, x_3)^T$ forms the required weighted combination of the three images.

Exercise 1.9.3: A vertical shear

The given images are $T(e_1) = e_1 - 3e_2 = (1, -3)^T$ and $T(e_2) = e_2 = (0, 1)^T$. Hence

$$A = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}, \qquad T(x_1, x_2) = (x_1, x_2 - 3x_1).$$

The horizontal coordinate stays fixed while the vertical coordinate changes in proportion to it.

Exercise 1.9.4: A horizontal shear

Here $T(e_1) = e_1$ and $T(e_2) = e_2 + 2e_1 = (2, 1)^T$. Thus

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \qquad T(x_1, x_2) = (x_1 + 2x_2, x_2).$$

This time the vertical coordinate stays fixed.

Exercise 1.9.5: A counterclockwise quarter turn

Rotation through $\pi/2$ takes e_1 to e_2 and e_2 to $-e_1$. Therefore

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The resulting rule $T(x_1, x_2) = (-x_2, x_1)$ agrees with both coordinate-vector images.

Exercise 1.9.6: A clockwise three-quarter turn

An angle of $-3\pi/2$ differs from $\pi/2$ by a full turn. Consequently the final coordinate-vector images are again $e_1 \mapsto e_2$ and $e_2 \mapsto -e_1$, giving

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The direction of travel is different, but the final transformation is the same.

Exercise 1.9.7: Rotate, then reflect

Rotation through $-3\pi/4$ has matrix

$$R = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}.$$

Reflection across the horizontal axis negates the second output coordinate, so it changes the signs in the second row of R. Thus

$$A = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}.$$

In particular the first column is $(-1/\sqrt{2}, 1/\sqrt{2})^T$, agreeing with the hint. The order is important: reflect the rotated image.

Exercise 1.9.8: Shear, then reflect across a sloping line

The shear is $S(x_1, x_2) = (x_1+2x_2, x_2)$. Reflection across $x_2 = -x_1$ takes (a, b) to $(-b, -a)$. Composing these two explicit rules gives

$$T(x_1, x_2) = (-x_2, -x_1 - 2x_2), \qquad A = \begin{bmatrix} 0 & -1 \\ -1 & -2 \end{bmatrix}.$$

The columns can also be checked by applying the two rules to e_1, e_2.

Exercise 1.9.9: Reflect, then rotate clockwise

The first step takes (x_1, x_2) to $(x_1, -x_2)$. A clockwise quarter turn takes (a, b) to $(b, -a)$. Hence the final image is $(-x_2, -x_1)$, and

$$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}.$$

Applying the two steps in the opposite order would produce a different map.

Exercise 1.9.10: Two reflections make a rotation

The horizontal reflection gives $(x_1, -x_2)$; reflection across $x_2 = x_1$ exchanges its coordinates. Thus

$$T(x_1, x_2) = (-x_2, x_1), \qquad A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The result is a counterclockwise quarter turn.

Exercise 1.9.11: Reflect across the two coordinate axes

After horizontal reflection, (x_1, x_2) becomes $(x_1, -x_2)$. Vertical reflection then gives $(-x_1, -x_2)$. The standard matrix is $-I_2$, whose columns are $-e_1, -e_2$. A rotation by π radians has the same two images, so it is the same linear transformation.

Conclusion. A half turn, through π radians (equivalently $-\pi$).

Exercise 1.9.12: Identify the angle from the coordinate-vector images

Exercise 10 gives $e_1 \mapsto e_2$, $e_2 \mapsto -e_1$. These are precisely the images under counterclockwise rotation by $\pi/2$. Since a linear map is determined by its values on the standard coordinate vectors, the two transformations agree on every vector.

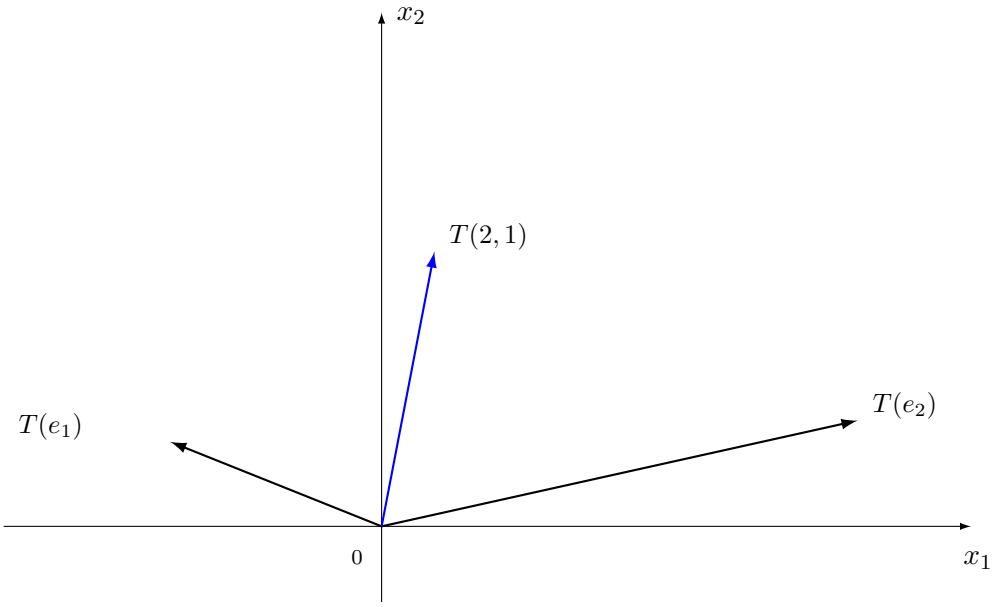
Conclusion. The rotation angle is $\pi/2$ modulo 2π.

Exercise 1.9.13: Draw a weighted sum of the two column images

The vector $(2, 1)^T$ equals $2e_1 + e_2$, so linearity gives

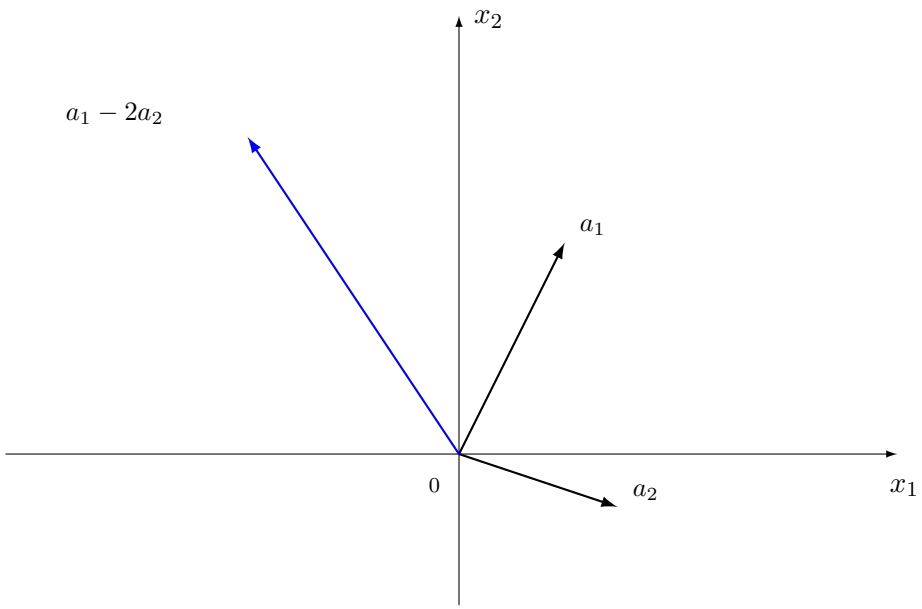
$$T(2, 1)^T = 2T(e_1) + T(e_2).$$

Take two copies of the up-left arrow $T(e_1)$, followed head to tail by one copy of the up-right arrow $T(e_2)$. The final arrow from zero ends high above the origin and slightly to its right. The following schematic preserves the source directions; the exact statement is the linear-combination formula, since the source gives no numerical coordinates for these image vectors.



Exercise 1.9.14: Subtract twice the second column

The product $A(1,-2)^T$ is $a_1 - 2a_2$. Reverse the up-right/down-right source arrow a_2 and double its length, then attach it at the tip of a_1. The resultant points up and to the left. This construction uses the original columns as vectors, not their lengths alone. A schematic is shown below.



Exercise 1.9.15: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 2 & -4 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 3 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} 2x_1 - 4x_2 \\ x_1 - x_3 \\ -x_2 + 3x_3 \end{bmatrix}.$$

The identity is required for *all* inputs, so these coefficients are forced; one numerical input would not be enough to determine all the entries.

Exercise 1.9.16: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 3 & -2 \\ 1 & 4 \\ 0 & 1 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} 3x_1 - 2x_2 \\ x_1 + 4x_2 \\ x_2 \end{bmatrix}.$$

The identity is required for *all* inputs, so these coefficients are forced; one numerical input would not be enough to determine all the entries.

Exercise 1.9.17: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} x_1 + 2x_2 \\ 0 \\ 2x_2 + x_4 \\ x_2 - x_4 \end{bmatrix}.$$

This is exactly the stated rule $T(x)$, so T is a matrix transformation and therefore linear. For any u, v, c, d, $A(cu + dv) = cAu + dAv$ verifies the definition.

Exercise 1.9.18: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 1 & 4 \\ 0 & 0 \\ 1 & -3 \\ 1 & 0 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} x_1 + 4x_2 \\ 0 \\ x_1 - 3x_2 \\ x_1 \end{bmatrix}.$$

This is exactly the stated rule $T(x)$, so T is a matrix transformation and therefore linear. For any u, v, c, d, $A(cu + dv) = cAu + dAv$ verifies the definition.

Exercise 1.9.19: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 1 & -5 & 4 \\ 0 & 1 & -6 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} x_1 - 5x_2 + 4x_3 \\ x_2 - 6x_3 \end{bmatrix}.$$

This is exactly the stated rule $T(x)$, so T is a matrix transformation and therefore linear. For any u, v, c, d, $A(cu + dv) = cAu + dAv$ verifies the definition.

Exercise 1.9.20: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 3 & 0 & 4 & -2 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} 3x_1 + 4x_3 - 2x_4 \end{bmatrix}.$$

This is exactly the stated rule $T(x)$, so T is a matrix transformation and therefore linear. For any u, v, c, d, $A(cu + dv) = cAu + dAv$ verifies the definition.

Exercise 1.9.21: Find the input producing two prescribed coordinates

The equations are $x_1 + x_2 = 3$ and $4x_1 + 5x_2 = 8$. Subtract four times the first from the second: $x_2 = -4$. Substitute into the first: $x_1 = 3 - (-4) = 7$. A check gives $T(7, -4) = (7 - 4, 28 - 20) = (3, 8)$.

Conclusion. $x = (7, -4)^T$. The elimination determines both variables uniquely.

Exercise 1.9.22: Check all three output coordinates

From $2x_1 - x_2 = 0$, obtain $x_2 = 2x_1$. The second equation $-3x_1 + x_2 = -1$ then becomes $-x_1 = -1$, so $x_1 = 1$ and $x_2 = 2$. The third equation must still be checked: $2(1) - 3(2) = -4$, as required.

Conclusion. $x = (1, 2)^T$. The third equation is consistent with the values forced by the first two.

Exercise 1.9.23: What coordinate-vector images determine

(a) **True.** Every $x \in \mathbb{R}^n$ is $\sum_j x_j e_j$, so linearity determines $T(x)$ from all the images $T(e_j)$.

(b) **True.** The rotation matrix $\begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$ implements the stated rotation, and multiplication by a matrix is linear.

(c) **False.** A composition of linear maps is always linear: $T(S(cu+dv)) = T(cS(u)+dS(v)) = cT(S(u)) + dT(S(v))$.

(d) **False.** Every map with that codomain sends each input somewhere in $\mathbb{R}^m$, whether or not it is onto. Onto means every *output* in $\mathbb{R}^m$ is reached by some input. The zero map to $\mathbb{R}$ is a counterexample to the claimed implication.

(e) **False.** The 3×2 matrix with columns e_1, e_2 in $\mathbb{R}^3$ has independent columns, so its map is one-to-one. Having more output coordinates than input coordinates does not obstruct injectivity.

Exercise 1.9.24: Distinguish being a function from being one-to-one

(a) **False.** A 4×3 matrix has at most three pivots, so it cannot have a pivot in all four rows. Its columns cannot span $\mathbb{R}^4$.

(b) **True.** The matrix with columns $T(e_j)$ implements any linear map between the stated coordinate spaces.

(c) **True.** This is exactly the column construction of the standard matrix.

(d) **False.** Having one output for each input merely says T is a function. One-to-one requires different inputs to give different outputs. The zero map has a unique output at every input but is not one-to-one on a nontrivial domain.

(e) **False.** A horizontal shear has matrix $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$, with nonzero shear amount k. The displayed diagonal matrices instead change signs of coordinate axes.

Exercise 1.9.25: An unused input coordinate and a zero output coordinate

For the map in Exercise 17, the third column is zero, so $T(e_3) = 0 = T(0)$ with $e_3 \neq 0$. Thus it is **not one-to-one**. The second output coordinate is always zero, so the vector $(0,1,0,0)^T$ is never attained. Therefore it is also **not onto** $\mathbb{R}^4$. These two failures concern different sides of the same matrix: a zero column creates nonuniqueness, and a zero row here prevents some outputs.

Exercise 1.9.26: A wide matrix can be onto but not one-to-one

The matrix in Exercise 2 is $\begin{bmatrix} 1 & -2 & 3 \\ 4 & 9 & -8 \end{bmatrix}$. Replacing R_2 by $R_2 - 4R_1$ gives $\begin{bmatrix} 1 & -2 & 3 \\ 0 & 17 & -20 \end{bmatrix}$. There is a pivot in both rows, so the map is **onto** $\mathbb{R}^2$. But three columns and only two pivots leave a free variable, so it is **not one-to-one**. For example $(-11, 20, 17)^T$ is a nonzero input sent to zero, as substitution verifies.

Exercise 1.9.27: Two pivot rows with one free variable

For Exercise 19, the matrix $\begin{bmatrix} 1 & -5 & 4 \\ 0 & 1 & -6 \end{bmatrix}$ is already in echelon form. Its two nonzero rows each have a pivot, making the map **onto** $\mathbb{R}^2$. The homogeneous equations yield $x_2 = 6x_3$ and $x_1 = 26x_3$, so $(26, 6, 1)^T$ is a nonzero input sent to zero. Hence the map is **not one-to-one**.

Exercise 1.9.28: Use the geometry of two column directions

In Exercise 14 the vectors a_1 and a_2 are nonzero and not parallel: the first points up-right and the second down-right. They are therefore independent, giving two pivot columns. In a 2×2 matrix those are also two pivot rows. Thus the map is **one-to-one and onto** $\mathbb{R}^2$. Equivalently, the two independent column directions span the whole plane, and each combination has unique coefficients.

Exercise 1.9.29: A tall one-to-one matrix

For $T : \mathbb{R}^3 \to \mathbb{R}^4$ to be one-to-one, its 4×3 matrix must have a pivot in every column. The sole echelon pattern is

$$\begin{bmatrix} p & * & * \\ 0 & q & * \\ 0 & 0 & r \\ 0 & 0 & 0 \end{bmatrix}, \qquad p, q, r \neq 0.$$

The last row must be zero because the three available pivots occupy the first three nonzero rows.

Exercise 1.9.30: Every full-row-rank three-by-four echelon pattern

Onto $\mathbb{R}^3$ requires three pivots, one in each row. They can occupy any three of the four columns, subject to moving right as one moves down. Thus the possible patterns are

$$\begin{bmatrix} p & * & * & * \\ 0 & q & * & * \\ 0 & 0 & r & * \end{bmatrix}, \qquad \begin{bmatrix} p & * & * & * \\ 0 & q & * & * \\ 0 & 0 & 0 & r \end{bmatrix},$$

$$\begin{bmatrix} p & * & * & * \\ 0 & 0 & q & * \\ 0 & 0 & 0 & r \end{bmatrix}, \qquad \begin{bmatrix} 0 & p & * & * \\ 0 & 0 & q & * \\ 0 & 0 & 0 & r \end{bmatrix}, \qquad p, q, r \neq 0.$$

Here $*$ denotes any entry. There are no additional patterns, because there are exactly four ways to choose three pivot columns out of four.

Exercise 1.9.31: Count columns for injectivity

The blank is $\boxed{n}$. One-to-one is equivalent to $Ax = 0$ having only zero as a solution. With n variables, this happens exactly when none is free, that is, when all n columns are pivot columns. Conversely any missing column pivot leaves a free variable and thus a nonzero input with the same image as zero.

Exercise 1.9.32: Count rows for surjectivity

The blank is $\boxed{m}$. Theorem 12 says onto is equivalent to columns spanning $\mathbb{R}^m$. Theorem 4 in Section 1.4 says this is equivalent to a pivot in every row. There are m rows, so there must be m pivots and hence m pivot columns. A row without a pivot would allow an inconsistent target after row reduction.

Exercise 1.9.33: The standard matrix is unique

Suppose $T(x) = Ax = Bx$ for every $x \in \mathbb{R}^n$. Substitute $x = e_j$. The vector Ae_j is column j of A and Be_j is column j of B. Both equal $T(e_j)$, so these columns agree for every $j = 1, \ldots, n$. Matrices with identical corresponding columns are identical, proving $A = B$. Equality on just a single nonbasis input would not imply equality of all columns.

Exercise 1.9.34: A composition preserves both operations

Take $u, v \in \mathbb{R}^p$ and scalars c, d. Linearity of S first gives $S(cu + dv) = cS(u) + dS(v)$, where the two image vectors lie in $\mathbb{R}^n$, the domain of T. Applying linearity of T then gives

$$T(S(cu + dv)) = T(cS(u) + dS(v)) = cT(S(u)) + dT(S(v)).$$

This is the combined linearity criterion for $T \circ S$ on $\mathbb{R}^p$, with outputs in $\mathbb{R}^m$. The domain matching is what makes the composition defined.

Exercise 1.9.35: Compare input and output dimensions

If T is onto, its $m \times n$ matrix has m pivot rows, so it needs at least m columns: $\boxed{m \leq n}$. If it is one-to-one, the matrix has n pivot columns and therefore needs at least n rows: $\boxed{n \leq m}$. If both properties hold, both inequalities force $m = n$. These are necessary dimension conditions, not sufficient ones; even a square zero matrix has neither property in positive dimension.

Exercise 1.9.36: Onto asks for the existence of a preimage

Saying T is onto means: for *every* b in the codomain, there *exists* an input x with $T(x) = b$. Thus it asks whether each target can be reached at all. It does not require that such an input be unique. A wide full-row-rank matrix can be onto while each target has infinitely many preimages.

Exercise 1.9.37: Use exact reduction to test the requested property

The source matrix and its exact reduced echelon form are

$$A = \begin{bmatrix} -5 & 6 & -5 & -6 \\ 8 & 3 & -3 & 8 \\ 2 & 9 & 5 & -12 \\ -3 & 2 & 7 & -12 \end{bmatrix},$$

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are only three pivots in four columns. Taking the free variable $x_4 = 1$ gives the nonzero solution $x = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ of $Ax = 0$. Direct multiplication verifies this relation. Hence T is not one-to-one.

Exercise 1.9.38: Use exact reduction to test the requested property

The source matrix and its exact reduced echelon form are

$$A = \begin{bmatrix} 7 & 5 & 9 & -9 \\ 5 & 6 & 4 & -4 \\ 4 & 8 & 0 & 7 \\ -6 & -6 & 6 & 5 \end{bmatrix},$$

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Every one of the four columns has a pivot, so $Ax = 0$ forces $x = 0$. Thus T is one-to-one. Since there is also a pivot in every row, this particular map is onto as well, although injectivity is the requested property.

Exercise 1.9.39: Use exact reduction to test the requested property

The source matrix and its exact reduced echelon form are

$$A = \begin{bmatrix} 4 & -7 & 3 & 7 & 5 \\ 6 & -8 & 5 & 12 & -8 \\ -7 & 10 & -8 & -9 & 14 \\ 3 & -5 & 4 & 2 & -6 \\ -5 & 6 & -6 & -7 & 3 \end{bmatrix},$$

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 5 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are four pivots and five rows. The zero row in the reduced coefficient matrix means there is not a pivot in every row, so the columns cannot span $\mathbb{R}^5$. Therefore T does not map $\mathbb{R}^5$ onto $\mathbb{R}^5$.

Exercise 1.9.40: Use exact reduction to test the requested property

The source matrix and its exact reduced echelon form are

$$A = \begin{bmatrix} 9 & 43 & 5 & 6 & -1 \\ 14 & 15 & -7 & -5 & 4 \\ -8 & -6 & 12 & -5 & -9 \\ -5 & -6 & -4 & 9 & 8 \\ 13 & 14 & 15 & 3 & 11 \end{bmatrix},$$

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

There is a pivot in every one of the five rows. Appending any right-hand side cannot create an inconsistent zero-coefficient row, so $Ax = b$ is solvable for every $b \in \mathbb{R}^5$. Thus T maps $\mathbb{R}^5$ onto $\mathbb{R}^5$; it is also one-to-one because every column has a pivot.